

1. Explain behaviour of hub device

HUB is used to connect multiple device it will share packet for all ^{hub device} (Broadcast device) it in layer 1 Physical layer hub it store the data opposite to opposite it will lost the packet name called (collision). HUB is a half duplex because once the person send the packet after that only another person can send.

2. Explain behaviour of switch device

Switch is used to connect multiple device. Switch have 8 port 10, 24, 48 Port A PC and switch connected when PC share in the packet to switch one process happen called (MAC table) ^(Switch layer 2 data link) Address learning ^{→ this process address learning} it will store the (Port No and Source mac) after that it will ^{may} send (Unicast) or else it will send it has Broadcast store it mac table.

Single broadcast domain → 3 group is connected to one switch it have more traffic (3 or more) broadcast domain → less traffic connection is not available to connect with wires act as single.

3. Explain behaviour of router device

Router is used to (break a broadcast device) it is used as layer 3 (Network layer) router act as unicast device. PC to router we are sharing a data means ^{to another PC} in which

Port you connect router e0, e1 you have to assign a IP address for router and then routing table store three things (network ID, Prefix, Interface) → this called as Directly connected routes. Find which PC to share and see the destination IP ^{checks} to routing table it has route means it send destination correctly or else drop the packet.